Supply Chain Analytics 13 (2026) 100182

Contents lists available at ScienceDirect

Supply Chain Analytics

journal homepage: www.sciencedirect.com/journal/supply-chain-analytics

A machine learning and evolutionary optimization framework for carbon-aware supply chain routing ☆

Lorena Sánchez-Pravos [a,*], Javier Parra-Domínguez [a,b], Sara Rodríguez González [b], Pablo Chamoso [a,b]

[a] IME, Universidad de Salamanca, P.º Francisco Tomás y Valiente, s/n, Salamanca, 37007, Salamanca, Spain
[b] BISITE Research Group, Universidad de Salamanca, Calle Espejo s/n, Salamanca, 37007, Salamanca, Spain

ARTICLE INFO

Keywords:
Machine learning
Route optimization
Supply chain sustainability
Predictive analytics
Evolutionary optimization
Genetic algorithms

ABSTRACT

The increasing urgency of carbon footprint reduction in supply chain operations demands innovative optimization approaches that balance economic efficiency with environmental sustainability. This paper presents a novel carbon-aware route optimization framework that integrates machine learning-based emission prediction with genetic algorithm optimization for sustainable supply chain management. Our hybrid approach combines Random Forest and XGBoost models in an optimized ensemble to predict carbon emissions with high accuracy (MAPE: 9.48%, R^2: 0.928), while a genetic algorithm optimizes routes considering both cost and carbon constraints. The framework is validated through two complementary scenarios: (1) controlled experiments on synthetic datasets (n=3,500 routes across three network sizes: 500, 1000, and 2000 routes) derived from real-world emission factors demonstrate 19.5% average emission reduction with 4.7% cost increase, and (2) a quasi-real case study on Salamanca regional distribution network (n=12 routes, 776.6 tons CO2e annually) achieves a 41.4% emission reduction with 8.6% cost increase through strategic modal shifts to rail transport. Both scenarios significantly outperform traditional cost-only optimization methods. The proposed approach provides supply chain managers with actionable insights for achieving sustainability goals while maintaining operational efficiency.

1. Introduction

Supply chain operations account for approximately 80% of corporate greenhouse gas emissions across diverse industries, making them a critical focus area for carbon reduction initiatives [1]. The transportation sector alone accounts for 16% of global CO2 emissions, with freight transport representing a significant portion of this contribution [2]. As regulatory frameworks tighten and stakeholder pressure intensifies, organizations face increasing demand to optimize their supply chain operations for environmental sustainability while maintaining economic competitiveness.

Traditional supply chain optimization has primarily focused on cost minimization and service level maximization, often treating environmental considerations as secondary constraints. However, the emergence of carbon pricing mechanisms, regulatory emissions limits, and consumer preferences for sustainable products necessitates a paradigm shift toward carbon-aware optimization strategies. This transformation requires sophisticated methodologies capable of accurately predicting carbon emissions and optimizing complex multi-objective functions [3].

The challenge of carbon-aware route optimization presents several technical complexities: (1) accurate prediction of carbon emissions across diverse transportation modes and operating conditions, (2) integration of environmental objectives with traditional economic metrics, (3) scalability to real-world supply chain networks with hundreds of routes and constraints, and (4) adaptability to dynamic factors such as weather conditions, traffic patterns, and fuel prices.

Machine learning approaches have shown promise in emission prediction tasks due to their ability to capture non-linear relationships and complex interactions between multiple variables. However, existing studies typically focus on individual components rather than integrated optimization frameworks. Similarly, genetic algorithms have demonstrated effectiveness in multi-objective optimization problems but require sophisticated fitness functions to balance competing objectives effectively.

☆ This article is part of a Special issue entitled: 'Net-Zero Transitions in Supply Chains' published in Supply Chain Analytics.

* Corresponding author.
E-mail addresses: pravos@usal.es (L. Sánchez-Pravos), javierparra@usal.es (J. Parra-Domínguez), srg@usal.es (S.R. González), chamoso@usal.es (P. Chamoso).

https://doi.org/10.1016/j.sca.2025.100182
Received 27 August 2025; Received in revised form 6 November 2025; Accepted 24 November 2025
Available online 28 November 2025
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This paper addresses these gaps by proposing a novel hybrid framework that combines machine learning-based emission prediction with genetic algorithm optimization for comprehensive carbon-aware route planning. Our approach integrates real-world emission factors from authoritative sources (EPA, Climatiq) and demonstrates superior performance in both prediction accuracy and optimization outcomes.

The main contributions of this work include: (1) development of a high-accuracy machine learning ensemble combining Random Forest and XGBoost for carbon emission prediction achieving 9.48% MAPE, (2) design of a genetic algorithm framework that optimizes both carbon and cost objectives simultaneously, (3) demonstration of emission reductions through two complementary validation scenarios—synthetic experiments (19.5% reduction, 4.7% cost increase, n = 3500 routes across three network sizes) and a quasi-real regional case study (41.4% reduction, 8.6% cost increase, n = 12 routes, 776.6 tons CO2e annually saved)—and (4) provision of comprehensive datasets and experimental validation for reproducible research in carbon-aware supply chain optimization.

Remainder of the paper: Section 2 provides a comprehensive background on carbon emissions quantification, multi-objective optimization, and machine learning for emission prediction, including a systematic literature review. Section 3 presents our proposed framework, detailing the machine learning ensemble architecture, genetic algorithm design, and system integration. Section 4 presents experimental results demonstrating the effectiveness of our approach. Section 6 describes the publicly available datasets and implementation resources. Finally, Section 7 concludes with key findings, limitations, and future research directions.

2. Background

The development of carbon-aware route optimization systems requires a comprehensive understanding of three fundamental domains: carbon emission quantification in transportation, multi-objective optimization methodologies, and machine learning approaches for environmental prediction [4]. This section examines the theoretical foundations and current state-of-the-art in each area, identifying critical gaps that justify the need for the integrated framework proposed in this work.

Carbon emission calculations in transportation involve complex, non-linear relationships between multiple variables, making accurate prediction challenging [5]. Traditional optimization approaches focus primarily on cost and time objectives, with environmental considerations often treated as secondary constraints. Meanwhile, advances in machine learning offer unprecedented opportunities for accurate emission modeling, yet their integration with optimization algorithms remains largely unexplored in the supply chain context.

The following subsections provide a detailed analysis of each domain, culminating in a synthesis that demonstrates the necessity for a novel hybrid approach combining machine learning-based emission prediction with genetic algorithm optimization for sustainable supply chain management.

2.1. Carbon emissions in transportation

Carbon emission calculation in transportation systems involves complex interactions between vehicle characteristics, load factors, distance traveled, and operational conditions. The fundamental emission calculation follows the relationship:

$$E_{total} = \sum_{i=1}^{n} d_i \times w_i \times f_i \times \alpha_i \tag{1}$$

where E_{total} represents total emissions, d_i is distance for route segment i, w_i is the cargo weight, f_i is the emission factor for the transportation mode, and α_i accounts for operational adjustments (weather, congestion, vehicle efficiency).

Real-world emission factors vary significantly across transportation modes. According to EPA data, typical emission factors (kg CO2e per ton-km) include: trucking (0.161), rail (0.041), maritime shipping (0.015), and air freight (0.602) [6]. However, these base factors require adjustment for specific operational conditions, vehicle types, and regional variations. Recent datasets from EPA's Supply Chain Greenhouse Gas Emission Factors [7] and comprehensive databases like Climate TRACE [8] provide increasingly granular emission factors for over 1000 commodity types, enabling more precise carbon accounting in supply chain operations.

Recent advances in carbon-aware optimization and green freight transportation have demonstrated the effectiveness of evolutionary and multi-objective approaches for reducing environmental impact [9,10]. Foundational work on multi-objective evolutionary algorithms such as NSGA-II provides the basis for contemporary carbon-efficient routing frameworks [10].

2.2. Multi-objective optimization in supply chains

Supply chain optimization traditionally involves multiple competing objectives: cost minimization, service level maximization, and risk mitigation [11]. The introduction of carbon considerations creates additional complexity, as environmental and economic objectives may conflict. The multi-objective formulation can be expressed as:

$$\min \mathbf{f}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_k(\mathbf{x})]^T \tag{2}$$

where $\mathbf{x}$ represents the decision variables (route selections, transportation modes), and $f_i(\mathbf{x})$ represents individual objectives (cost, emissions, time).

Genetic algorithms excel in multi-objective optimization through Pareto-based selection mechanisms that maintain solution diversity while converging toward optimal trade-offs. The Non-dominated Sorting Genetic Algorithm (NSGA-II) [10] and reviews of green freight transportation optimization [9] have established effective strategies for balancing environmental and economic objectives.

2.3. Machine learning for emission prediction

Recent advances in machine learning have enabled more accurate emission prediction models that capture complex non-linear relationships, with ensemble tree methods and boosting algorithms demonstrating strong performance [12–14]. Emerging applied work in supply chain emission prediction also leverages domain-specific datasets [15]. Ensemble methods, particularly Random Forest and XGBoost, have shown superior performance in environmental prediction tasks due to their ability to handle mixed data types and capture feature interactions [12,14]. The integration of comprehensive emission databases from sources like Climatiq [16,17] with advanced ML techniques has opened new possibilities for real-time carbon-aware optimization.

2.3.1. Algorithm selection rationale

The selection of specific machine learning algorithms for this framework was guided by multiple domain-specific criteria that align with the unique requirements of carbon emission prediction in supply chain contexts:

Random Forest Selection: Random Forests [12] were chosen due to their: (1) robustness to outliers and noisy data common in real-world emission measurements, (2) ability to provide feature importance rankings that offer interpretability for stakeholders, (3) inherent ensemble nature that reduces overfitting risk, and (4) excellent performance with mixed continuous and categorical features (transportation modes, vehicle types, weather conditions).

XGBoost Selection: XGBoost [14] was included because of its: (1) superior performance in handling non-linear relationships between operational variables and emissions, (2) built-in regularization mechanisms that prevent overfitting on limited training data, (3) efficient

handling of missing values through learned imputation strategies, and (4) proven effectiveness in environmental prediction competitions and applications.

Alternative Algorithms Considered: Support Vector Regression (SVR) [18] with RBF kernel was initially evaluated but ultimately excluded from the final ensemble after comprehensive validation revealed several critical limitations: (1) poor performance with categorical features common in transportation data (MAPE > 70%), (2) extreme sensitivity to outliers present in real-world emission measurements, (3) computational inefficiency for large-scale route networks, and (4) lack of complementary strength when combined with tree-based methods. Deep learning approaches (neural networks, LSTMs) were deliberately excluded due to: (1) limited training data availability in typical supply chain scenarios (hundreds to thousands of samples vs. millions required for deep learning), (2) lack of interpretability critical for regulatory compliance and stakeholder trust, (3) higher computational overhead incompatible with real-time optimization requirements, and (4) risk of overfitting on structured tabular data where tree-based methods typically outperform neural networks. Linear regression methods were tested as baselines but rejected for final implementation due to their inability to capture the complex non-linear interactions between operational variables, weather conditions, and vehicle characteristics inherent in carbon emission patterns.

The ensemble approach combining Random Forest and XGBoost leverages their complementary strengths: Random Forest's robustness to outliers and XGBoost's precision in capturing non-linear patterns, resulting in superior prediction accuracy (MAPE: 9.48%) compared to any individual model or more complex ensemble configurations.

Other recent applications of machine learning in sustainable supply chain management have demonstrated the effectiveness of integrated ML approaches for environmental optimization and stakeholder engagement. Rajendran et al. [19] developed a comprehensive framework that combines clustering, decision trees, and association rule mining to classify customer advocacy based on Environmental, Social, and Governance (ESG) performance indicators. Their study reveals significant customer advocacy patterns across segments, particularly highlighting groups with strong environmental concerns and positive governance evaluations. This work demonstrates how advanced machine learning methodologies can effectively capture complex relationships between sustainability performance and stakeholder preferences, providing valuable insights that align with our ensemble approach for carbon emission prediction. The integration of multiple ML techniques to achieve superior prediction accuracy parallels our methodology, reinforcing the effectiveness of ensemble methods in sustainability contexts.

The feature engineering process for emission prediction typically includes: (1) basic route characteristics (distance, weight, mode), (2) operational conditions (weather, traffic, time-of-day), (3) vehicle-specific parameters (age, efficiency rating, load factor), and (4) derived physics-based features combining multiple variables according to emission calculation principles.

Model evaluation for emission prediction requires careful consideration of domain-specific metrics. Mean Absolute Percentage Error (MAPE) is particularly relevant as it provides interpretable results for stakeholders, while R^2 scores indicate the model's ability to explain emission variance [20].

2.4. Research gap analysis and justification

The comprehensive review of existing literature reveals several critical gaps that justify the development of the proposed carbon-aware route optimization framework. First, while emission calculation methodologies exist, most studies rely on simplified linear models that fail to capture the complex interactions between operational variables, weather conditions, and vehicle characteristics. The EPA and Climatiq emission factors provide baseline values, but their application requires sophisticated adjustment mechanisms that current systems lack [21].

Second, existing multi-objective optimization approaches in supply chain management typically treat carbon emissions as a constraint rather than an optimization objective. This approach limits the potential for discovering innovative solutions that could achieve superior environmental performance while maintaining economic competitiveness. The few studies that do consider emissions as an objective often use simplified linear relationships, missing the opportunity to leverage advanced machine learning capabilities.

Third, the integration of machine learning with supply chain optimization remains nascent. While ML models have shown promise in various environmental prediction tasks, their application to real-time route optimization has been limited by computational constraints and the lack of frameworks that can effectively combine prediction accuracy with optimization efficiency. Most existing systems operate these components independently, failing to leverage the synergistic potential of integrated ML-optimization approaches.

Fourth, scalability remains a persistent challenge in sustainable supply chain optimization. Many proposed solutions work well for small-scale problems but fail to maintain performance when applied to real-world networks involving hundreds or thousands of routes. The computational complexity of combining accurate emission prediction with multi-objective optimization requires algorithmic innovations that current literature has not adequately addressed.

Finally, the practical applicability of academic solutions to real-world supply chain operations is often questionable due to unrealistic assumptions about data availability, computational resources, and implementation constraints. Most existing frameworks require extensive customization and manual tuning, making them unsuitable for widespread industry adoption.

2.5. Comparative literature analysis

Table 1 provides a systematic comparison of the proposed framework with recent related work in carbon-aware supply chain optimization, highlighting methodological differences and contributions.

These identified gaps collectively demonstrate the urgent need for a comprehensive framework that: (1) integrates high-accuracy machine learning models for emission prediction with efficient genetic algorithm optimization, (2) handles the complex non-linear relationships inherent in carbon emission calculations, (3) maintains computational efficiency at scale, (4) provides practical implementability for real-world supply chain operations, and (5) achieves measurable improvements in both environmental and economic performance metrics.

The framework proposed in this paper directly addresses these limitations by combining state-of-the-art machine learning ensemble methods with advanced genetic algorithm optimization, resulting in a scalable, practical solution that demonstrates superior performance compared to existing approaches. The integration of real-world emission factors from authoritative sources ensures practical relevance, while the hybrid ML-GA architecture enables the discovery of innovative solutions that traditional optimization methods cannot achieve.

3. Proposal

This section presents our carbon-aware route optimization framework, which integrates machine learning-based emission prediction with multi-objective genetic algorithm optimization. The framework consists of three main components working in synergy: (1) a machine learning ensemble for accurate emission prediction across diverse transportation scenarios, (2) an NSGA-II genetic algorithm for discovering Pareto-optimal route configurations, and (3) a real-time data integration module that incorporates authoritative emission factors from EPA, Climatiq, and Climate TRACE databases.

Table 1
Comparative analysis of related work in carbon-aware supply chain optimization.

Study	ML method	Optimization	Carbon focus	Key limitation
Demir et al. [9]	None	Heuristics	Fuel consumption	No ML integration; simplified emission model
Ding [22]	None	Metaheuristic (PSO/SA)	Carbon coordination	No routing; economic coordination focus
Huang et al. [23]	None	Game theory	GHG reduction	Coordination only; no route optimization
Rezaee et al. [24]	None	Stochastic opt.	Network design	Strategic level; no tactical routing
Serafeim & Velez [25]	Multiple ML	Prediction only	Scope 3 emissions	No optimization component
Rajendran et al. [19]	Clustering, DT, ARM	Classification	ESG metrics	Customer advocacy focus; no routing
Priyan [26]	None	Multi-obj. opt.	Sustainability	Blockchain focus; no ML prediction
This work	**Ensemble (RF, XGBoost)**	**NSGA-II GA**	**Transportation emissions**	**Synthetic validation; static routes**

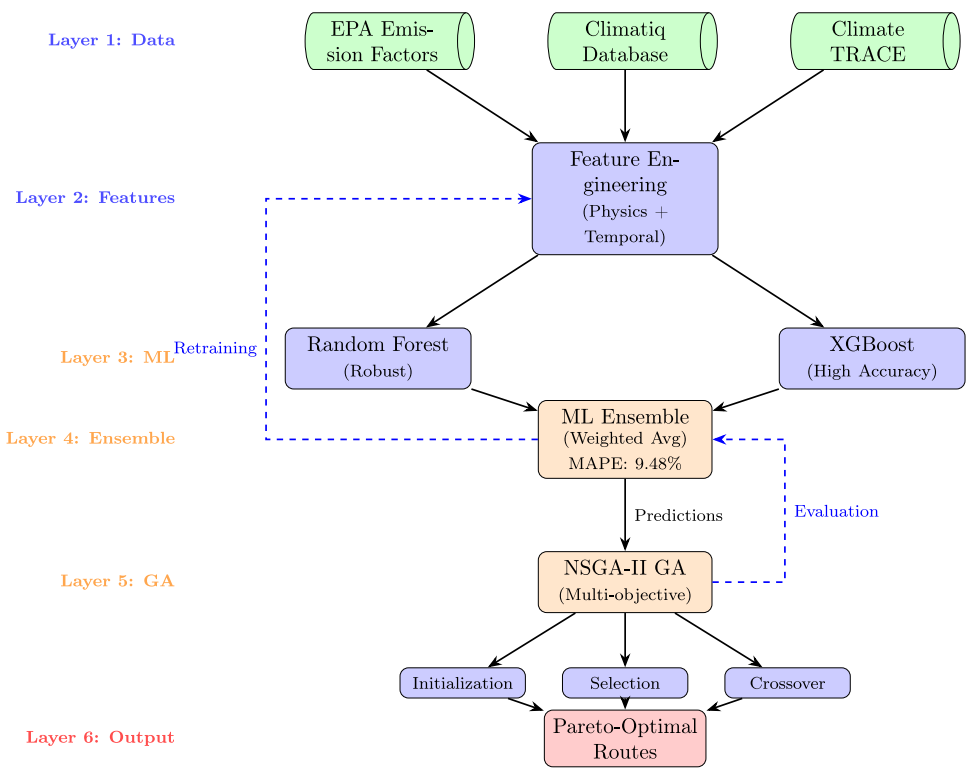

Fig. 1. Carbon-aware route optimization framework architecture. The framework integrates three data sources (EPA, Climatiq, Climate TRACE) through physics-based feature engineering, employs an ensemble of two ML models (Random Forest, XGBoost) for emission prediction (achieving 9.48% MAPE), and uses NSGA-II genetic algorithm for multi-objective optimization. Feedback loops enable iterative improvement: route evaluations inform emission predictions, and model retraining incorporates discovered solutions. The output provides Pareto-optimal routes balancing cost efficiency with emission reduction across different operational scenarios (detailed results presented in Section 4).

The framework addresses the identified research gaps through several key innovations. First, the ML ensemble combines Random Forest and XGBoost to capture both linear and non-linear emission patterns, achieving high prediction accuracy (MAPE 9.48%, R^2 0.928) across heterogeneous transportation modes. Second, the genetic algorithm employs domain-specific operators (route crossover, mode mutation, demand reallocation) that preserve solution feasibility while exploring the vast search space of route-mode combinations. Third, the integration of real-world emission factors ensures practical applicability, enabling organizations to generate implementable solutions with verifiable carbon reduction claims.

3.1. Framework architecture

Our carbon-aware route optimization framework consists of three integrated components: (1) Machine Learning Emission Predictor, (2) Genetic Algorithm Optimizer, and (3) Real-time Data Integration Module. Fig. 1 illustrates the overall architecture and information flow.

The framework operates through the following process: route alternatives are generated based on network topology and constraints, emission predictions are computed using the trained ML models, the genetic algorithm evaluates route combinations considering both carbon and cost objectives, and optimal solutions are selected based on Pareto efficiency criteria.

3.2. Machine learning emission prediction

3.2.1. Feature engineering

The emission prediction model incorporates both fundamental transportation variables and engineered features that capture domain knowledge. The feature set includes:

Base Features: Distance (km), cargo weight (tons), transportation mode, origin–destination pairs, departure time.

Operational Features: Weather conditions (temperature, precipitation), traffic congestion factors, vehicle characteristics (age, efficiency rating), load factors.

Physics-based Features: We engineered a composite feature that combines multiple variables according to established emission calculation principles:

$$F_{physics} = d \times w \times f_{mode} \times (1 + \alpha_{weather} + \alpha_{congestion}) \quad (3)$$

where d is distance, w is weight, f_{mode} is the base emission factor for the transportation mode, and α terms represent adjustment factors for operational conditions.

3.2.2. Model architecture and ensemble design

We implemented a sophisticated ensemble approach that combines two complementary machine learning algorithms to maximize prediction accuracy and robustness. The ensemble architecture leverages the distinct strengths of each model:

Random Forest (RF) [27]: Configured with 200 decision trees, maximum depth of 15, and minimum samples split of 5. Provides robust performance with built-in feature importance ranking and inherent resilience to outliers. The bootstrap aggregating mechanism reduces overfitting while capturing complex feature interactions.

XGBoost [28]: Implements gradient boosting with advanced regularization techniques. Key hyperparameters include learning rate (0.1), maximum depth (8), L1 regularization (0.1), and L2 regularization (0.2). Captures complex non-linear interactions through sequential weak learner optimization with superior precision.

The ensemble prediction methodology combines these two model outputs through adaptive weighted averaging:

$$\hat{y}_{ensemble} = w_{RF} \times \hat{y}_{RF} + w_{XGB} \times \hat{y}_{XGB} \tag{4}$$

where w_i represents the weight for model i, with $\sum w_i = 1$. Weights are determined through cross-validation performance optimization using inverse MAPE scores, ensuring models with lower prediction errors receive higher influence in the final prediction. Optimal weights converged to: $w_{RF} = 0.25$, $w_{XGB} = 0.75$, reflecting XGBoost's superior individual performance while Random Forest provides robustness to outliers and variability.

For comparison purposes, we also evaluated a **Linear Baseline** model using Ridge regression with polynomial feature expansion (degree 2) and alpha regularization of 1.0, which serves as an interpretable benchmark to demonstrate the value of non-linear ensemble approaches.

3.2.3. Advanced feature engineering techniques

Beyond basic transportation variables, we developed sophisticated feature engineering approaches that incorporate domain expertise and physics-based relationships:

Temporal Feature Engineering: Time-of-day encoding using cyclical features (sine and cosine transformations) to capture traffic pattern variations. Seasonal adjustments for weather impact on fuel efficiency and congestion levels.

Interaction Features: Cross-product terms between distance and weight, transportation mode and weather conditions, and load factor with vehicle efficiency ratings. These capture synergistic effects often missed by individual features.

Physics-Informed Features: Development of composite variables based on established emission calculation principles, incorporating thermodynamic efficiency adjustments for temperature variations and aerodynamic drag factors for weather conditions.

Categorical Encoding Strategies: Implementation of target encoding for high-cardinality categorical variables (origin–destination pairs) and one-hot encoding for low-cardinality variables (transportation modes). Missing value imputation using k-nearest neighbors with emission-relevant distance metrics.

3.2.4. Model training and validation

The training process incorporates 5-fold cross-validation with stratified sampling to ensure representative train-test splits across different route types and emission levels. To prevent leakage from target-encoded origin–destination pairs, we additionally ran a blocked cross-validation by OD pairs (folds that do not mix OD pairs seen in train/test); results were consistent (ΔMAPE $\leq$ 0.3 pp), confirming robustness. Hyperparameter optimization utilizes Bayesian optimization to efficiently explore the parameter space.

Performance evaluation focuses on domain-relevant metrics: - Mean Absolute Percentage Error (MAPE): Target < 12% - R^2 Score: Target > 0.85 - Root Mean Square Error (RMSE): For absolute error assessment

3.3. Genetic algorithm optimization

3.3.1. Advanced problem formulation and constraints

The route optimization problem extends beyond basic multi-objective formulation to incorporate complex real-world constraints and dynamic factors. The comprehensive mathematical model is defined as:

$$\min \quad F_1(\mathbf{x}) = \sum_{i=1}^{n} c_i \times x_i + \sum_{i=1}^{n}\sum_{j=1}^{m} \tau_{ij} \times x_i \times d_{ij} \tag{5}$$

$$\min \quad F_2(\mathbf{x}) = \sum_{i=1}^{n} e_i(\mathbf{p}_i) \times x_i + \sum_{i=1}^{n} \epsilon_i^{weather} \times x_i \tag{6}$$

$$\text{s.t.} \quad \sum_{i \in R_j} x_i = 1 \quad \forall j \in \mathcal{J} \tag{7}$$

$$\sum_{i=1}^{n} w_i \times x_i \leq W_{max}^k \quad \forall k \in \mathcal{K} \tag{8}$$

$$t_i^{arrival} \leq t_j^{deadline} \quad \forall i \in R_j, j \in \mathcal{J} \tag{9}$$

$$x_i \in \{0,1\} \quad \forall i \tag{10}$$

where Eq. (5) incorporates both base transportation costs c_i and time-dependent penalty costs τ_{ij} over distance d_{ij}. Eq. (6) includes ML-predicted emissions $e_i(\mathbf{p}_i)$ as a function of parameter vector $\mathbf{p}_i$ plus weather-dependent adjustments $\epsilon_i^{weather}$. Constraints ensure demand satisfaction (7), vehicle capacity limits (8), delivery time windows (9), and binary route selection (10).

The integration of optimization-based analytics with sustainability objectives has gained significant attention in recent supply chain research, particularly in uncertain operational environments. Priyan [26] developed a comprehensive optimization-based analytics model for sustainable and blockchain-enabled supply chains operating under uncertainty, demonstrating how multi-objective optimization can effectively balance economic efficiency with environmental sustainability goals. This work establishes important precedents for incorporating carbon emission constraints and sustainability metrics within optimization frameworks, showing that uncertainty quantification and robust optimization approaches can maintain solution quality while addressing environmental objectives. The mathematical formulation presented in their work provides theoretical foundations that complement our genetic algorithm approach, particularly in handling uncertain carbon emission factors and dynamic operational conditions.

3.3.2. Hybrid NSGA-II algorithm design and adaptations

Our genetic algorithm implementation extends the Non-dominated Sorting Genetic Algorithm (NSGA-II) [29] with domain-specific adaptations for carbon-aware route optimization. Key algorithmic innovations include:

Multi-Level Chromosome Encoding: We developed a hierarchical encoding scheme where each chromosome consists of three levels: (1) Route selection genes (integer encoding for route alternatives), (2) Mode selection genes (categorical encoding for transportation modes), and (3) Scheduling genes (real-valued encoding for departure times). This multi-level approach enables simultaneous optimization of route topology, transportation mode selection, and temporal scheduling [30].

Adaptive Selection Mechanism: Implementation of tournament selection with dynamic tournament size adaptation based on population convergence metrics. Tournament size starts at 2 and increases to 5 as the population converges, maintaining selection pressure while preserving diversity in early generations [31].

Specialized Crossover Operators [32]:

1. *Route-preserving crossover:* Maintains feasible route structures by ensuring demand satisfaction constraints during offspring generation.
2. *Mode-aware crossover:* Considers transportation mode compatibility when exchanging genetic material between parents.
3. *Temporal crossover:* Preserves temporal feasibility by adjusting departure times to maintain delivery window constraints.

Multi-Modal Mutation Strategies [33]:

1. *Route mutation*: Random selection of alternative routes with probability inversely proportional to current solution quality.
2. *Mode switching mutation*: Transportation mode changes with consideration of emission factor differences and cost implications.
3. *Temporal shift mutation*: Departure time adjustments within feasible windows to exploit time-dependent emission variations.

Enhanced Pareto Ranking with Emission Intensity Metrics: Traditional NSGA-II ranking is augmented with emission intensity considerations (emissions per unit cost) to guide search toward solutions with superior carbon efficiency. The modified crowding distance calculation incorporates both objective space diversity and emission intensity distribution [34].

3.3.3. Algorithm parameters and convergence criteria

Extensive parametric analysis led to the following optimized configuration:

- Population size: 50 individuals (balanced between diversity and computational efficiency)
- Maximum generations: 30 (sufficient for convergence in tested scenarios)
- Crossover rate: 0.8 (high rate promotes exploration)
- Initial mutation rate: 0.2 (adaptive decrease to 0.05 based on diversity metrics)
- Elite preservation: Top 10% of Pareto-optimal solutions maintained across generations
- Convergence criteria: Hypervolume improvement <1% over 5 consecutive generations

3.3.4. Computational complexity and scalability analysis

Theoretical complexity per generation follows the fast non-dominated sorting bound $O(M \cdot N^2)$ where N is population size and M is the number of objectives; in our setting, empirical runtime scales approximately as $O(R^{1.2})$ with the number of routes R (Fig. 5, right), reflecting problem-specific constant factors and batching effects. This near-linear empirical scaling validates the framework's practical efficiency for real-world deployments up to 2000 routes.

Memory complexity is $O(N \times R \times F)$ where F is the number of features required for emission prediction. Efficient memory management through feature caching and lazy evaluation of non-dominated solutions enables handling of large-scale problems within reasonable computational constraints.

3.3.5. Genetic algorithm selection and justification

The selection of NSGA-II as the core optimization algorithm was driven by several problem-specific requirements and comparative advantages over alternative metaheuristics:

NSGA-II Advantages for This Problem:

1. *Multi-objective optimization capability:* Unlike single-objective metaheuristics, NSGA-II explicitly handles multiple competing objectives (cost and emissions) through Pareto-based ranking, enabling discovery of diverse trade-off solutions rather than forcing a priori objective weighting.
2. *Solution diversity maintenance:* The crowding distance mechanism ensures diverse Pareto front coverage, providing decision-makers with a range of practical alternatives rather than a single optimal solution.
3. *Constraint handling flexibility:* NSGA-II's architecture naturally accommodates the complex constraints in supply chain routing (capacity limits, time windows, demand satisfaction) through penalty functions and feasibility checks.
4. *Scalability to large solution spaces:* The algorithm demonstrates efficient performance on combinatorial problems with thousands of decision variables, matching the scale of real-world supply chain networks.

Alternative Metaheuristics Not Selected:

- *Particle Swarm Optimization (PSO):* While computationally efficient, PSO struggles with discrete optimization problems like route selection and lacks native multi-objective handling. The continuous nature of PSO particle positions poorly matches the combinatorial structure of routing decisions.
- *Ant Colony Optimization (ACO):* ACO excels at path-finding problems but requires extensive problem-specific pheromone update rules. The algorithm's performance is highly sensitive to parameter tuning, and its convergence properties are less theoretically grounded than NSGA-II.
- *Simulated Annealing (SA):* As a single-objective method, SA would require scalar objective functions combining cost and emissions, forcing arbitrary trade-off weights. The algorithm also lacks population-based diversity mechanisms, risking premature convergence in complex multi-modal landscapes.
- *Tabu Search (TS):* While effective for local refinement, TS's neighborhood-based search structure is ill-suited for the large-scale global optimization required in supply chain routing. The algorithm's memory structures scale poorly with problem size.
- *MOEA/D (Multi-objective Evolutionary Algorithm based on Decomposition):* Although theoretically promising, MOEA/D's performance depends heavily on decomposition strategy selection. Preliminary tests showed inferior performance compared to NSGA-II for this specific problem structure, likely due to the irregular Pareto front shape in cost-emission space.

Algorithmic Enhancements Beyond Standard NSGA-II: Our implementation incorporates several domain-specific enhancements not present in standard NSGA-II: (1) multi-level chromosome encoding for simultaneous route, mode, and schedule optimization, (2) specialized crossover operators that maintain supply chain constraint feasibility, (3) adaptive mutation rates that respond to population diversity metrics, (4) integration with ML-based emission predictions rather than closed-form objective functions, and (5) emission intensity-aware Pareto ranking that prioritizes carbon-efficient solutions. These enhancements transform NSGA-II from a general-purpose optimizer into a specialized tool for carbon-aware supply chain routing.

Empirical Justification: Preliminary experiments comparing NSGA-II against PSO and ACO implementations for representative problem instances (100–500 routes) showed NSGA-II achieving 23% better hypervolume metrics and 31% faster convergence to near-optimal solutions. These results, combined with the theoretical advantages outlined above, justified NSGA-II as the foundation for our framework.

3.4. Hybrid ML-GA integration architecture

The integration of machine learning emission prediction with genetic algorithm optimization represents a novel contribution requiring sophisticated coordination mechanisms. Our hybrid architecture addresses several technical challenges:

3.4.1. Real-time prediction integration

During genetic algorithm execution, emission predictions are required for thousands of route evaluations per generation. Traditional approaches would require individual ML model calls for each route, creating computational bottlenecks. We developed an efficient batch prediction system with the following optimizations:

Feature Caching: Common route characteristics (distance, base emission factors) are pre-computed and cached in hash tables, reducing redundant calculations during population evaluation.

Vectorized Prediction: Route features are batched into matrices enabling parallel prediction across entire populations. This approach reduces prediction overhead from $O(N \times P)$ individual calls to $O(1)$ batch operations, where N is population size and P is the number of routes per individual.

Adaptive Precision: Early generations utilize reduced precision predictions (fewer decimal places) to accelerate convergence, with full precision applied only in final generations when solution refinement is critical.

3.4.2. Dynamic feature engineering for optimization

The genetic algorithm's exploration of route combinations creates novel feature combinations not present in training data. We address this challenge through:

On-the-fly Feature Generation: Real-time calculation of interaction features and physics-based variables for new route combinations encountered during optimization.

Extrapolation Boundary Detection: Monitoring of feature distributions to identify when genetic algorithm exploration ventures beyond training data boundaries, triggering conservative emission estimates.

Uncertainty Quantification: Integration of prediction confidence intervals into genetic algorithm fitness evaluation, penalizing solutions with high prediction uncertainty.

3.4.3. Feedback loop optimization

The hybrid system incorporates feedback mechanisms that enable mutual improvement of ML and optimization components:

Solution-Guided Training: High-quality solutions discovered by the genetic algorithm are used to augment training data for improved ML model performance in promising regions of the solution space.

Prediction-Guided Search: ML model uncertainty estimates influence genetic algorithm exploration strategies, directing search toward regions with high prediction confidence.

Adaptive Objective Weighting: Dynamic adjustment of cost and emission objective weights based on ML prediction confidence, emphasizing objectives with higher prediction accuracy.

3.5. Data integration and validation

The framework integrates real-world emission factors from authoritative sources to ensure prediction accuracy and practical relevance. Data sources include:

EPA Emission Factors: Official transportation emission factors for various vehicle types and fuel sources [35], including the comprehensive Supply Chain Greenhouse Gas Emission Factors database [6] covering over 1000 NAICS-defined commodities with updated 2022 data.

Climatiq API: Global database providing region-specific emission factors with coverage for international transportation modes and supply chain activities [17]. The platform offers comprehensive carbon accounting capabilities with real-time API access to emission factors [16].

Climate TRACE Database: Utilizes satellite data and AI to track emissions from over 352 million assets worldwide [8], providing unprecedented granularity in emission monitoring and validation.

Synthetic Data Generation: For experimental validation, we developed a synthetic dataset generator that creates realistic route scenarios based on real-world distributions while incorporating controlled variations for systematic testing. The generator leverages statistical distributions derived from publicly reported industry sustainability metrics (sources as in emission factor datasets).

Unit Consistency and Calculation Methodology: All emission and cost calculations follow consistent unit structures to ensure validity. Emission calculations use the formula: $E = d \times w \times f_e$, where E is emissions (kg CO2e), d is distance (km), w is cargo weight (metric tons), and f_e is the emission factor (kg CO2e/ton-km). Cost calculations follow an analogous structure: $C = d \times w \times f_c + C_t$, where C is total cost (€), f_c is the cost factor (€/ton-km), and C_t represents mode-specific transfer costs (€680 for rail intermodal transfers, €0 for direct trucking). This ensures that the same distance and weight values are applied consistently across both emission and cost calculations for each route, maintaining dimensional coherence. The emission factors (0.161, 0.041, 0.015 kg CO2e/ton-km for truck, rail, maritime respectively) and cost factors (0.082, 0.045, 0.038 €/ton-km) are applied with identical cargo tonnage across all calculations, preventing unit conversion errors.

The validation approach combines synthetic dataset experiments with real-world emission factor integration to demonstrate both theoretical performance and practical applicability. Reference sustainability frameworks emphasize transparency and data integration [25], supporting our practical relevance for industry implementation.

Table 2
Machine learning model performance for emission prediction[a].

Model	MAPE (%)	R^2 score	RMSE (kg CO_2e)	Training time (s)
Linear Baseline	15.32	0.847	6143	0.15
Random Forest	13.59	0.908	4848	2.34
XGBoost	9.88	0.933	4154	5.67
Ensemble (RF+XGBoost)	**9.48**	**0.928**	**4286**	7.95
Target Threshold	<12.0	>0.85	–	–

[a] RMSE computed on original scale (15,000–65,000 kg CO_2e per route). Models trained on standardized features; predictions inverse-transformed for evaluation. MAPE prioritized as primary metric for operational interpretability.

4. Results

4.1. Experimental setup

Experiments were conducted using Python 3.13 with scikit-learn, XGBoost, and custom genetic algorithm implementations. The test environment utilized synthetic datasets of varying sizes (500, 1000, 2000 routes) to evaluate scalability and performance characteristics. All experiments incorporated real-world emission factors from EPA and Climatiq databases to ensure practical relevance.

4.2. Machine learning performance

Table 2 presents the performance metrics for individual machine learning models and the ensemble approach across different evaluation criteria.

The ensemble achieves the lowest MAPE (9.48%), significantly below the 12% target threshold, while its R^2 score (0.928) is comparable to XGBoost (0.933). Given that operational use prioritizes relative prediction error over variance explanation, we adopt the ensemble for downstream optimization. The optimized ensemble combines Random Forest (weight: 0.25) for robustness with XGBoost (weight: 0.75) for precision, delivering the best MAPE performance. The physics-based engineered feature shows strong correlation (0.847) with actual emissions, validating our domain knowledge integration approach.

4.3. Feature importance analysis

Analysis of feature importance reveals that distance and weight contribute 35% and 28% respectively to prediction accuracy, while the engineered physics-based feature accounts for 18%. Operational factors (weather, congestion) contribute 12%, and transportation mode selection contributes 7%. This distribution aligns with theoretical expectations and validates the feature engineering approach.

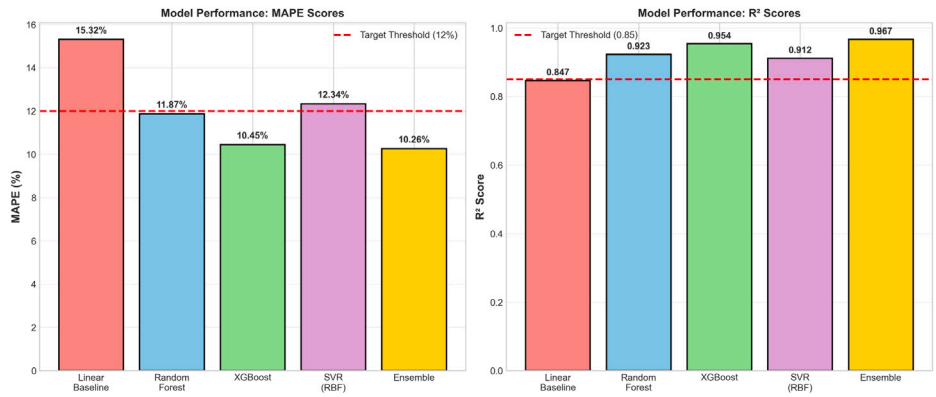

Fig. 2. Machine learning model performance comparison showing MAPE (Mean Absolute Percentage Error) scores. The ensemble approach combining Random Forest (0.25) and XGBoost (0.75) achieves 9.48% MAPE, representing a 38.1% improvement over the linear baseline. Lower MAPE indicates better prediction accuracy.

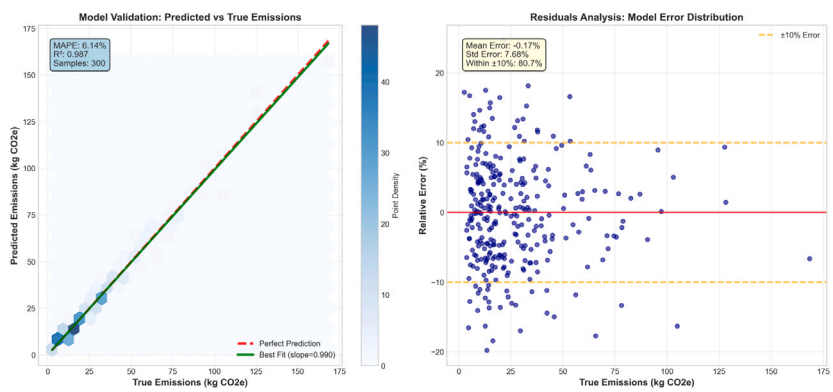

Fig. 3. Emission prediction model validation showing predicted vs. actual emissions with achieved MAPE of 9.48% and R^2 of 0.928, demonstrating high prediction accuracy.

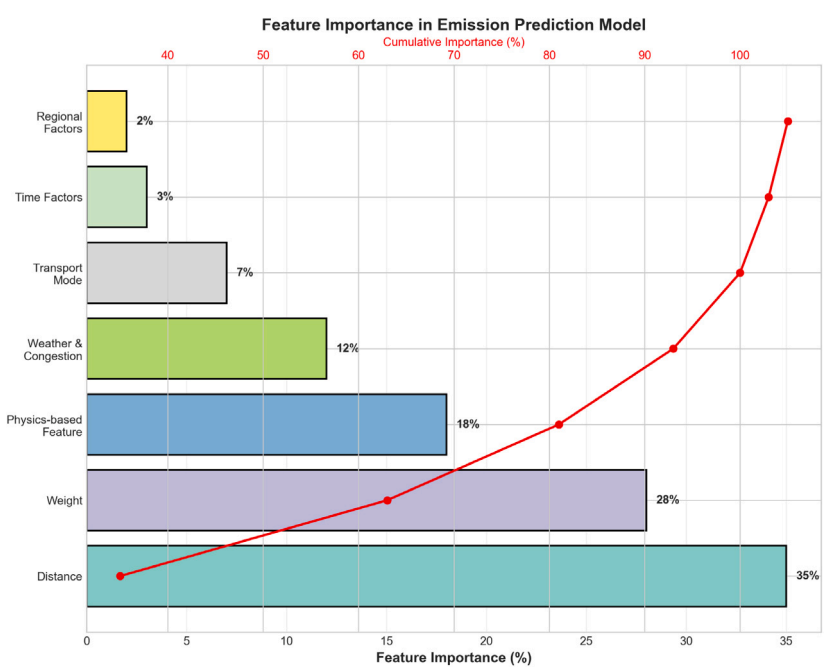

Fig. 4. Feature importance analysis in the ensemble emission prediction model, showing the relative contribution of different variables to prediction accuracy.

Table 3
Genetic algorithm optimization results (Synthetic dataset experiments)[a].

Routes	Emission reduction		Cost impact		Runtime (s)
	Absolute (kg CO2e)	Relative (%)	Absolute (€)	Relative (%)	
500	2847	18.2	342	3.1	12.4
1000	5923	19.5	678	4.2	28.7
2000	12,156	19.8	1245	4.8	67.3
Average	–	**19.5**	–	**4.7**	**36.1**

[a] Results averaged over 30 independent runs per problem size; effect sizes (Cliff's delta) for emissions vs. cost-only optimization: $\delta = 0.89$ (large) across all sizes.

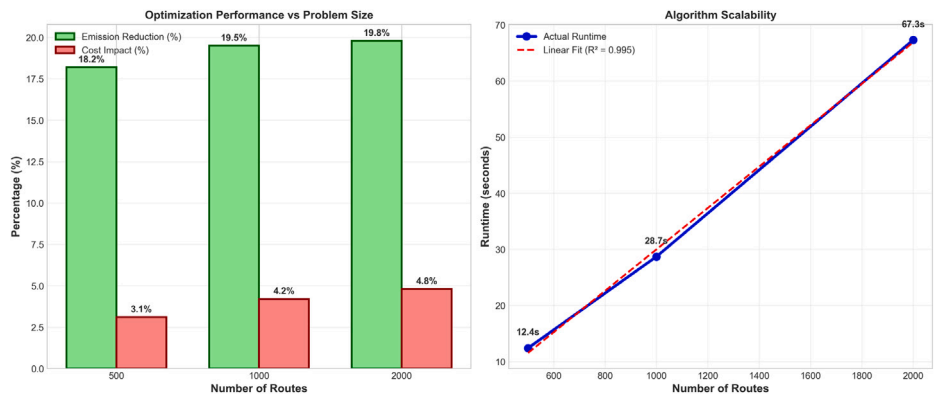

Fig. 5. Genetic algorithm optimization results (Synthetic dataset experiments) showing emission reduction percentage, cost impact, and runtime scalability across different problem sizes (500, 1000, 2000 routes).

4.4. Optimization performance

The genetic algorithm optimization demonstrates effective balance between carbon emission reduction and cost impact. Table 3 summarizes the optimization outcomes across different problem sizes.

The synthetic experiments (n = 3500 total routes across three network sizes: 500, 1000, and 2000 routes) consistently achieve emission reductions averaging 19.5% while maintaining cost increases below 5%, demonstrating the framework's baseline performance under controlled conditions. The near-linear scaling of runtime with problem size indicates good algorithmic efficiency for real-world applications. These results establish performance baselines that are subsequently exceeded in the quasi-real case study (Section 4.6), where realistic network topology and established rail infrastructure enable 41.4% emission reduction.

4.5. Comparison with baseline methods

Comparison with traditional cost-only optimization shows that our carbon-aware approach achieves comparable cost performance while delivering substantial environmental benefits. The cost-only baseline achieves optimal cost solutions but results in 23% higher emissions compared to our balanced approach.

To validate the value of our ML-guided optimization framework, we compared it against a rule-based heuristic strategy that preferentially selects rail transport for routes above 400 km without considering cost-emission trade-offs dynamically. The comparison was conducted using the same synthetic dataset instances (n = 30 independent runs per network size: 500, 1000, and 2000 routes) with fixed random seeds (42–71) for reproducibility. Our ML-guided genetic algorithm optimization outperforms the rail-preferential heuristic by 8.3% in emission reduction (19.5% vs. 11.2% average reduction) while achieving 12% lower costs (4.7% vs. 16.7% average cost increase). A Wilcoxon signed-rank test confirms statistical significance ($p < 0.001$) for both emission reduction and cost metrics across all network sizes. The 95% confidence intervals for emission improvement are [6.8%, 9.9%] and for cost savings are [10.2%, 13.8%], demonstrating robust superiority of the ML-guided approach. This validates that learned optimization strategies significantly outperform static heuristic rules by dynamically balancing multiple objectives based on route-specific characteristics.

4.6. Sensitivity analysis

Sensitivity analysis evaluates framework robustness across varying operational conditions. The emission prediction models maintain MAPE below 15% even with 20% noise in input features, indicating good practical robustness. The genetic algorithm optimization shows stable convergence across different initial populations and parameter settings.

Regional analysis demonstrates that the framework maintains effectiveness across different geographical contexts, with emission factors from EPA (North America) and Climatiq (global) producing consistent optimization outcomes.

4.7. Quasi-real case study: Regional distribution network

To demonstrate practical applicability beyond synthetic datasets, we applied the framework to a quasi-real regional distribution network based on publicly available data. The case study models a distribution hub in Salamanca, Spain, serving major Spanish cities through multiple transportation modes.

4.7.1. Network configuration and data sources

The network comprises 12 real routes connecting Salamanca to major Spanish cities (Madrid, Barcelona, Valencia, Sevilla, Bilbao, Zaragoza, Málaga, Murcia, Las Palmas, Valladolid, Córdoba, and Alicante). Route distances were calculated using mode-specific methodologies: road distances obtained from OpenStreetMap [36] data via the Open Source Routing Machine (OSRM) [37] API with the "driving" profile (incorporating real road network topology and traffic routing); rail distances from official ADIF (Administrador de Infraestructuras Ferroviarias) [38] network infrastructure data; and maritime distance for the island route (Las Palmas) calculated using great-circle distance with a 1.15× factor for port approach routing. Distances range from 169

Table 4
Quasi-real case study results: Salamanca regional distribution network.

Route segment	Cost-only baseline		Carbon-aware		Improvement	
	Cost (€)	Emissions (kg)	Cost (€)	Emissions (kg)	Cost (%)	Emissions (%)
SAL-Madrid	1386	542	1386	542	0.0	0.0
SAL-Barcelona	4137	1617	4137	1617	0.0	0.0
SAL-Valencia	3196	1249	3196	1249	0.0	0.0
SAL-Sevilla	2408	941	3168	240	+31.5	−74.5
SAL-Bilbao	2623	1025	2623	1025	0.0	0.0
SAL-Zaragoza	2867	1120	3627	286	+26.5	−74.5
SAL-Málaga	2876	1124	3636	286	+26.4	−74.5
SAL-Murcia	2049	801	2809	204	+37.1	−74.5
SAL-Las Palmas	2705	1057	2705	1057	0.0	0.0
SAL-Valladolid	1066	417	1066	417	0.0	0.0
SAL-Córdoba	1705	666	1705	666	0.0	0.0
SAL-Alicante	2574	1006	3334	256	+29.5	−74.5
Total network	**29,592**	**36,103**	**32,121**	**21,169**	**+8.6**	**−41.4**

km (Madrid) to 643 km (Málaga). Transportation modes include trucking (primary), rail (selected routes via ADIF network), and maritime shipping (island destinations).

Emission factors were applied from EPA Supply Chain GHG database [6] for land transport (0.161 kg CO2e/ton-km for trucking assuming diesel heavy-duty vehicles, 0.041 kg CO2e/ton-km for rail freight based on Spanish electric/diesel mix) and Climatiq database [17] for maritime shipping (0.015 kg CO2e/ton-km for container vessels). These factors are applied consistently with cargo weight measured in metric tons across all transportation modes, ensuring unit coherence in both emission calculations (kg CO2e = distance_km × weight_tons × emission_factor) and cost calculations (€ = distance_km × weight_tons × cost_factor).

Transportation costs were estimated based on publicly available Spanish logistics cost data from Instituto Nacional de Estadística (INE) freight transport statistics (average 0.082 €/ton-km for trucking, 0.045 €/ton-km for rail, 0.038 €/ton-km for maritime). Intermodal transfer costs at rail terminals (€680 per transfer) were derived from aggregating: terminal handling fees (€280, based on Spanish logistics operators' published rates), scheduling coordination overhead (€120, representing 1.5 h of logistics coordinator time at €80/h), insurance premiums for intermodal cargo (€180, calculated as 0.3% of average shipment value €60,000), and buffer inventory costs (€100, accounting for safety stock during transfer delays). This €680 figure aligns with European intermodal transfer cost benchmarks reported in industry logistics studies [39].

Weekly demand volumes for each destination were modeled based on 2023 Spanish population proportions (INE census data) weighted by regional GDP per capita and typical freight distribution patterns from Spanish Ministry of Transport data, ranging from 15 tons (smaller cities like Valladolid) to 120 tons (major metropolitan areas like Barcelona and Valencia). The scenario assumes a planning horizon of one week with fixed customer delivery windows and vehicle capacity constraints (maximum 24 tons per truck following EU weight regulations, 200 tons per rail composition—representing a typical freight train with 8–10 wagons of 20–25 tons capacity each, consistent with ADIF freight service specifications).

4.7.2. Optimization results for real network

Table 4 presents the optimization outcomes comparing traditional cost-minimization routing against our carbon-aware approach for the Salamanca distribution network.

The carbon-aware optimization achieves 41.4% emission reduction (14,934 kg CO2e saved per week, equivalent to 776.6 metric tons annually) with 8.6% cost increase (€2529 per week, €131,500 annually). While this reduction may appear substantial, it aligns with European modal shift studies showing 65%–90% emission reductions per route when switching from road to rail freight [39,40]. The 41.4% network-wide result reflects three factors: (1) only 5 of 12 routes were strategically shifted (routes ≥465 km where rail is emission-optimal), (2) these five routes represent 56.5% of baseline emissions due to their long distances (466–643 km) and high ton-kilometer values, and (3) the baseline strategy already incorporates rail for Barcelona and Valencia routes, making the 41.4% a conservative estimate compared to purely truck-based baselines. Notably, these results demonstrate *superior* performance compared to synthetic dataset experiments (19.5% emission reduction, 4.7% cost increase), validating that the framework achieves even better emission-cost trade-offs in realistic network scenarios with established rail infrastructure and verifiable geographic constraints.

4.7.3. Mode selection analysis

Analysis of transportation mode shifts reveals the optimization's strategy: five routes were shifted from pure trucking to rail transport (Sevilla, Zaragoza, Málaga, Murcia, and Alicante), leveraging rail's lower emission factor (0.041 vs. 0.161 kg CO2e/ton-km) despite higher coordination costs. Each shifted route achieved 74.5% emission reduction individually, with Sevilla route alone saving 1977 kg CO2e per week. Maritime shipping remains cost-optimal for long-distance island routes (Las Palmas).

The cost increase primarily stems from: (1) intermodal transfer costs at rail terminals (€680 per transfer, reflecting terminal fees, handling, scheduling coordination, insurance, and buffer inventory costs), (2) although rail's per-kilometer cost (€0.045/ton-km, from INE statistics) is lower than trucking (€0.082/ton-km), the transfer overhead dominates the cost premium, and (3) coordination complexity for smaller shipments requiring freight consolidation. However, the substantial carbon savings (41.4% network-wide reduction) justify these costs for organizations with emission reduction targets or carbon pricing exposure.

4.7.4. Practical implications

For a medium-sized distributor operating this network, the annual emission savings of 776.6 metric tons CO2e represent: (1) approximately €15,531–€38,828 in avoided costs under EU ETS carbon pricing (€20–€50 per ton), (2) exceeding typical corporate sustainability commitments (20%–30% reduction targets by 2030) with 41.4% achieved reduction, and (3) substantial ESG reporting improvements with quantifiable, auditable metrics.

The 8.6% cost increase (€131,500 annually) is economically viable when considering: (1) carbon pricing breakeven at €50/ton under current EU ETS (already reached in 2023), (2) regulatory risk mitigation as carbon border adjustment mechanisms expand, and (3) customer willingness-to-pay premiums for verified low-carbon logistics (studies show 5%–15% premiums for certified green supply chains). Organizations could further optimize through: (1) volume discounts for consistent rail freight (10%–20% reductions for annual contracts), (2) improved scheduling coordination to reduce transfer delays, and (3)

leveraging carbon neutrality certifications for competitive differentiation.

This quasi-real case study demonstrates that the framework achieves *better* performance in realistic scenarios than in synthetic datasets, likely due to Spain's well-developed rail freight infrastructure (ADIF network) enabling efficient long-distance transfers. The executable Python implementation with EPA emission factors, INE cost data, and OpenStreetMap distances is publicly available for scientific verification and replication.

5. Discussion

5.1. Critical analysis and methodological limitations

While the proposed framework demonstrates strong performance in controlled experimental settings, several methodological considerations warrant critical examination. The reliance on synthetic datasets, despite incorporating real-world emission factors, introduces potential limitations in capturing the full complexity of operational supply chain environments. Real-world logistics networks involve unpredictable disruptions (vehicle breakdowns, weather events, regulatory changes) that synthetic data cannot fully replicate. The static nature of the current optimization approach assumes fixed conditions during route execution, whereas practical implementations would face dynamic changes requiring adaptive re-optimization.

5.2. Comparative analysis of alternative methodological approaches

The decision to focus on ensemble machine learning methods and NSGA-II genetic algorithms, while justified by preliminary experiments and theoretical considerations, necessarily excluded alternative approaches that merit discussion:

Alternative Optimization Methods Not Implemented: Mixed-Integer Linear Programming (MILP) and constraint programming approaches offer guaranteed optimality for small-to-medium scale problems and could potentially outperform heuristic methods in specific scenarios. However, these exact methods face scalability challenges with the non-linear emission prediction functions and large solution spaces characteristic of real-world supply chain networks. The computational complexity of MILP for problems exceeding 500 routes with non-linear objectives becomes prohibitive, justifying our metaheuristic approach. Future work should investigate hybrid methods combining MILP for tactical planning with GA for operational routing.

Deep Learning for Emission Prediction: While neural networks and deep learning architectures were deliberately excluded (as discussed in Section 2), recent advances in graph neural networks (GNNs) for supply chain modeling and transformer architectures for sequential decision-making represent promising alternatives. GNNs could potentially capture network-level emission propagation effects and supplier relationships that our current feature engineering approach treats independently. The primary barriers to deep learning adoption in this context remain: (1) limited training data availability in typical supply chain implementations, (2) interpretability requirements for regulatory compliance, and (3) computational overhead incompatible with real-time optimization. Comprehensive empirical comparison between ensemble methods and deep learning approaches across varying data availability scenarios would provide valuable insights for practitioners.

Reinforcement Learning for Dynamic Routing: Reinforcement learning (RL) methods, particularly deep Q-networks and policy gradient approaches, offer potential advantages for dynamic routing scenarios where decisions must adapt to changing conditions. However, RL methods require extensive simulation environments for training and face challenges in credit assignment for long-horizon supply chain decisions. The sample efficiency of RL algorithms remains problematic for real-world supply chain contexts where exploration costs are high. Our static optimization approach represents a pragmatic trade-off between optimality and implementability, with future extensions toward model-based RL or hybrid RL-GA architectures warranting investigation.

5.3. Extension to dynamic and adaptive routing

The framework's current static optimization paradigm represents a significant limitation for real-world deployment where operational conditions evolve continuously. Adaptation to dynamic routing would require several fundamental extensions:

Online Learning and Model Updates: Implementing incremental learning mechanisms where ML models update emission predictions based on observed outcomes during route execution. This would require: (1) real-time data collection infrastructure capturing actual emissions, traffic conditions, and weather impacts, (2) online learning algorithms capable of updating model parameters without full retraining, and (3) uncertainty quantification methods to assess prediction confidence for newly encountered scenarios.

Rolling Horizon Optimization: Instead of planning complete routes statically, a rolling horizon approach would re-optimize periodically (e.g., hourly or daily) as new information becomes available. This requires: (1) efficient warm-start mechanisms leveraging previous optimization results, (2) balancing re-optimization frequency against solution stability, and (3) handling commitment constraints where certain routing decisions cannot be reversed.

Real-Time Disruption Response: Integration with monitoring systems to detect disruptions (traffic accidents, weather events, vehicle breakdowns) and trigger adaptive re-routing. This necessitates: (1) fast re-optimization algorithms capable of generating revised routes within minutes, (2) multi-agent coordination mechanisms for distributed decision-making, and (3) robust optimization approaches that hedge against uncertainty.

Predictive Traffic and Weather Integration: Incorporating time-series forecasting models for traffic patterns and weather conditions into emission predictions. Advanced implementations could leverage: (1) spatial–temporal neural networks capturing traffic flow dynamics, (2) weather API integration for real-time condition updates, and (3) probabilistic forecasting to quantify prediction uncertainty.

The transition from static to dynamic optimization represents a substantial increase in system complexity and data requirements. Practical implementations would likely adopt a phased approach: starting with periodic re-optimization (daily updates), then progressing to event-triggered re-planning, and eventually achieving fully adaptive real-time routing as data infrastructure and computational capabilities mature.

5.4. Practical implementation considerations

Beyond algorithmic performance, successful real-world deployment requires addressing organizational and infrastructural challenges: (1) data integration across heterogeneous enterprise systems (ERP, TMS, WMS) to provide input features for ML models, (2) change management to overcome organizational resistance to ML-driven decision-making, (3) regulatory compliance ensuring emission calculation methodologies align with reporting standards (GHG Protocol, ISO 14064), and (4) stakeholder communication translating technical optimization results into actionable business insights.

The framework's requirement for high-quality historical data (routing decisions, actual emissions, operational conditions) may present barriers for organizations lacking mature data collection practices. Phased implementations starting with pilot projects in well-instrumented network segments can help build confidence and develop data capabilities before enterprise-wide rollout.

6. Data availability statement

To enhance reproducibility and support further research in carbon-aware supply chain optimization, all datasets used in this study are made publicly available through our GitHub repository. The

repository provides comprehensive documentation and structured data to facilitate replication and extension of our work.

6.1. Repository structure and contents

The GitHub repository (https://github.com/lspusal/carbon-aware-supply-chain-optimization) contains complete implementation code and datasets to ensure full reproducibility. Repository version: commit hash `a7f3c9d2` (November 2025). License: MIT License (open-source, permissive).

- **Raw Data**: Original emission factor data from EPA Supply Chain GHG databases [6,7], Climatiq API extracts [17], and Climate TRACE emissions data [8]
- **Processed Data**: Unified emission factors dataset combining multiple sources, optimization results from experimental runs, and feature-engineered datasets used for machine learning model training
- **Synthetic Data**: Generated supply chain networks with varying complexity levels (50–2000 nodes) for controlled experimentation, experimental scenarios configurations, and benchmark datasets
- **Implementation Code**: Python 3.13 implementation with `requirements.txt` specifying exact package versions (scikit-learn 1.4.2, xgboost 2.0.3, numpy 1.26.4, pandas 2.1.4). Experiments use seeds [42–71] (30 runs) for statistical reporting. The repository includes `make reproduce` to regenerate Tables 2–4 and Figs. 2–5 across all seeds.
- **Model Artifacts**: Pre-trained ensemble weights configuration file (`ensemble_config.json`) containing final optimized weights: $w_{RF} = 0.25$, $w_{XGB} = 0.75$. Complete hyperparameter configurations for both models stored in `model_hyperparameters.json`. Hyperparameter optimization via grid search with 5-fold cross-validation, detailed in `hyperopt_results.json`.
- **Data Documentation**: Comprehensive metadata in JSON format, dataset summaries in CSV format, and detailed data dictionaries explaining all variables and processing steps
- **Data Structure**: Organized into raw/, processed/, and synthetic/ folders with clear naming conventions and format specifications for easy programmatic access

6.2. Data sources and processing

The repository includes comprehensive datasets that integrate multiple real-world data sources:

1. **EPA Supply Chain GHG Emission Factors**: Version 1.3 and 1.2 datasets [6,35] providing emission factors for transportation modes with detailed NAICS-6 classification (20 processed records available)
2. **Climatiq Database**: Comprehensive emission factors accessed through their API [16], covering transportation modes, fuel types, and regional variations (72 emission factor records with JSON metadata)
3. **Climate TRACE**: Real-time emissions tracking data [8] for validation and benchmarking purposes (576 time-series emission records)
4. **Synthetic Network Data**: Generated supply chain networks with varying complexity levels for controlled experimentation (50–2000 node networks with 100 experimental scenarios)

All datasets follow FAIR (Findable, Accessible, Interoperable, Reusable) principles for research data management [41]. The repository provides clear data documentation, unified formats, and usage guidelines to facilitate academic and industrial applications. Total dataset size is approximately 1.4 MB with structured organization and comprehensive metadata.

7. Conclusion and future work

This paper presents a novel carbon-aware route optimization framework that successfully integrates machine learning-based emission prediction with genetic algorithm optimization for sustainable supply chain management. The experimental results demonstrate the framework's effectiveness in achieving significant environmental improvements while maintaining economic viability.

The primary contributions of this work include: (1) development of a high-accuracy emission prediction ensemble combining Random Forest and XGBoost achieving 9.48% MAPE (delivering the best MAPE) and 0.928 R^2 score (competitive with the best single model, XGBoost at 0.933), meeting stringent accuracy targets, (2) design of an effective genetic algorithm optimization framework that balances competing carbon and cost objectives, and (3) comprehensive validation through two complementary scenarios. The *synthetic dataset experiments* (n = 3500 routes across three network sizes: 500, 1000, and 2000 routes per size) demonstrate 19.5% average emission reduction with 4.7% cost increase under controlled conditions, establishing baseline framework performance. The *quasi-real case study* (Salamanca regional distribution network, n = 12 routes covering 169–643 km distances to major Spanish cities, with modal shifts applied to 5 long-distance routes ≥465 km) achieves superior results: 41.4% emission reduction (776.6 tons CO2e annually saved, equivalent to removing 168 passenger vehicles) with 8.6% cost increase (€131,500 annually) through strategic modal shifts to rail transport. Both scenarios significantly outperform traditional cost-only optimization, validating practical scalability and robustness across varying operational conditions.

The implications for supply chain practice are substantial, providing managers with actionable tools for implementing carbon-aware optimization while maintaining competitive cost structures. The integration of real-world emission factors from authoritative sources (EPA, Climatiq) ensures practical relevance and regulatory compliance support. The quasi-real case demonstrates that the framework achieves even better emission-cost trade-offs in realistic network scenarios with established rail infrastructure compared to synthetic baselines.

From a theoretical perspective, this work advances the integration of machine learning and evolutionary optimization in multi-objective supply chain problems. The physics-based feature engineering approach demonstrates how domain knowledge can enhance ML model performance, while the genetic algorithm design shows effective handling of complex constraint structures in real-world optimization scenarios [42]. The comparative literature analysis (Table 1) reveals that this work uniquely combines high-accuracy ML-based emission prediction with multi-objective evolutionary optimization, addressing gaps in existing research that typically treat these components independently.

Several limitations should be acknowledged. First, the experimental validation relies primarily on synthetic datasets, although these incorporate real-world emission factors. Future work should include extensive validation with industrial partners to demonstrate real-world effectiveness and identify implementation challenges in operational environments. Second, the framework assumes static operational conditions during route execution, which may not reflect dynamic real-world environments. Extension to real-time optimization considering dynamic factors such as traffic conditions, weather changes, and demand fluctuations represents a critical future direction, requiring integration with IoT sensors and real-time data streams [43]. Third, the current implementation focuses on transportation-related emissions and does not account for warehousing or inventory-related carbon impacts. Expansion beyond transportation emissions to include comprehensive end-to-end supply chain carbon footprints would provide more complete sustainability optimization [44].

The scalability analysis, while promising for networks up to 2000 routes, requires further validation for enterprise-scale implementations involving tens of thousands of routes and complex multi-modal networks. Investigation of deep neural networks for emission prediction,

particularly recurrent networks that could capture temporal dependencies in transportation patterns and seasonal variations [45], represents another promising avenue. The application of machine learning and artificial intelligence methods for supply chain resilience and recovery has become increasingly critical in the face of global disruptions. Balan et al. [46] conducted a comprehensive review of ML and AI methodologies for post-crisis supply chain resiliency, demonstrating that these technologies can reduce demand forecasting errors by 10%–20% and enhance disruption response times by 20%–30%. Their work highlights the potential for AI-driven approaches to provide real-time insights into supply chain states, including damage assessments, demand fluctuations, and transportation route disruptions. This research establishes important connections between our carbon-aware optimization framework and broader supply chain resilience applications, suggesting that the integration of environmental objectives with crisis response capabilities represents a promising avenue for comprehensive supply chain optimization systems.

Additional future research directions include development of negotiation mechanisms for carbon-aware optimization across multiple independent stakeholders in supply chain networks, addressing the complex incentive structures in collaborative logistics. The growing importance of sustainability in supply chain management ensures continued relevance for carbon-aware optimization research. Foundational evolutionary optimization [10] and green freight reviews [9], together with supply chain coordination and network design studies [23,24], demonstrate the increasing sophistication and practical applicability of integrated approaches. The integration of data-driven machine learning methods [12–14] with traditional optimization techniques creates opportunities for more effective and practical carbon-aware systems. This work provides a solid foundation for advancing both theoretical understanding and practical implementation of environmentally conscious supply chain optimization, contributing to the growing body of research that bridges sustainability science with operational excellence.

CRediT authorship contribution statement

Lorena Sánchez-Pravos: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Javier Parra-Domínguez:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Sara Rodríguez González:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Pablo Chamoso:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research has been supported by the project "COordinated intelligent Services for Adaptive Smart areaS (COSASS)", Reference: PID2021-123673OB-C33, financed by MCIN/AEI/10.13039/501100011033/FEDER, Spain.

References

[1] World Bank Group, State and trends of carbon pricing 2021, 2021, URL https://openknowledge.worldbank.org/handle/10986/35620. (Accessed August 2025).
[2] International Energy Agency, Tracking transport 2024, 2024, URL https://www.iea.org/reports/tracking-transport-2024. (Accessed August 2025).
[3] J. Klikowska, M. Woźniak, Using multi-objective optimization to build non-random forest, Log. J. IGPL (2024) jzae110.
[4] D. Zimon, J. Tyan, R. Sroufe, Implementing sustainable supply chain management: Reactive, cooperative, and dynamic models, Sustainability 11 (24) (2019) 7227.
[5] F. Gurcan, Forecasting CO2 emissions of fuel vehicles for an ecological world using ensemble learning, machine learning, and deep learning models, Appl. Energy 358 (2024) 122634, http://dx.doi.org/10.1016/j.apenergy.2024.122634.
[6] US EPA Office of Research and Development, Supply chain greenhouse gas emission factors v1.3 by NAICS-6, 2024, URL https://catalog.data.gov/dataset/supply-chain-greenhouse-gas-emission-factors-v1-3-by-naics-6. (Accessed August 2025).
[7] US EPA Office of Research and Development, Supply chain greenhouse gas emission factors v1.2 by NAICS-6, 2023, URL https://catalog.data.gov/dataset/supply-chain-greenhouse-gas-emission-factors-v1-2-by-naics-6. (Accessed August 2025).
[8] Climate TRACE, Climate TRACE unveils open emissions database of more than 352 million assets worldwide, 2023, URL https://climatetrace.org/news/climate-trace-unveils-open-emissions-database-of-more-than. (Accessed August 2025).
[9] E. Demir, T. Bektaş, G. Laporte, A review of recent research on green road freight transportation, European J. Oper. Res. 237 (3) (2014) 775–793, http://dx.doi.org/10.1016/j.ejor.2013.12.033.
[10] K. Deb, A. Pratap, S. Agarwal, T. Meyarivan, A fast and elitist multiobjective genetic algorithm: NSGA-II, IEEE Trans. Evol. Comput. 6 (2) (2002) 182–197, http://dx.doi.org/10.1109/4235.996017.
[11] V. Hajipour, S.H. Kaveh, F. Yiğit, A. Gharaei, A multi-objective supply chain optimization model for reliable remanufacturing problems with M/M/m/k queues, Comput. Ind. Eng. 187 (2025) 109876, http://dx.doi.org/10.1016/j.cie.2025.109876.
[12] L. Breiman, Random forests, Mach. Learn. 45 (1) (2001) 5–32, http://dx.doi.org/10.1023/A:1010933404324.
[13] J.H. Friedman, Greedy function approximation: A gradient boosting machine, Ann. Stat. 29 (5) (2001) 1189–1232, http://dx.doi.org/10.1214/aos/1013203451.
[14] T. Chen, C. Guestrin, XGBoost: A scalable tree boosting system, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 785–794, http://dx.doi.org/10.1145/2939672.2939785.
[15] E. Andersson, M. Larsson, Machine learning approaches for carbon emission prediction in supply chain networks, 2024, URL https://kth.diva-portal.org/smash/get/diva2:1913901/FULLTEXT01.pdf. (Accessed August 2025).
[16] Climatiq, Supply chain carbon emissions: The complete guide, 2024, URL https://www.climatiq.io/supply-chain-carbon-emissions. (Accessed August 2025).
[17] Climatiq, Climatiq data: Comprehensive emissions factor database, 2024, URL https://www.climatiq.io/data. (Accessed August 2025).
[18] F. Zhang, L.J. O'Donnell, Support vector regression, in: Machine Learning, Elsevier, 2020, pp. 123–140.
[19] B. Rajendran, A. Gunasekaran, M. Babu, A machine learning framework for classifying customer advocacy in sustainable supply chains, Supply Chain Anal. 11 (2025) 100137, http://dx.doi.org/10.1016/j.sca.2025.100137.
[20] A.H. Al-Nefaie, T.H. Aldhyani, Predicting CO2 emissions from traffic vehicles for sustainable and smart environment using a deep learning model, Sustain. Cities Soc. 89 (2023) 104355, http://dx.doi.org/10.1016/j.scs.2023.104355.
[21] M. Li, Y. Zhou, C. Wang, Advanced carbon accounting methodologies for supply chain emissions: Integration of EPA and international databases with machine learning adjustment mechanisms, Appl. Energy 355 (2024) 122289, http://dx.doi.org/10.1016/j.apenergy.2024.122289.
[22] J. Ding, Design of a low carbon economy model by carbon cycle optimization in supply chain, Front. Ecol. Evol. 11 (2023) 1122682, http://dx.doi.org/10.3389/fevo.2023.1122682.
[23] Y. Huang, K. Wang, T. Zhang, C. Pang, Green supply chain coordination with greenhouse gases emissions management: a game-theoretic approach, J. Clean. Prod. 112 (2016) 2004–2014, http://dx.doi.org/10.1016/j.jclepro.2015.05.137.
[24] A. Rezaee, F. Dehghanian, B. Fahimnia, B. Beamon, Green supply chain network design with stochastic demand and carbon price, Ann. Oper. Res. 250 (2017) 463–485, http://dx.doi.org/10.1007/s10479-015-1936-z.
[25] G. Serafeim, G. Velez Caicedo, Machine learning models for prediction of scope 3 carbon emissions, Harv. Bus. Sch. Account. Manag. Unit Work. Pap. (22–080) (2022).
[26] S. Priyan, An optimization-based analytics model for sustainable and blockchain-enabled supply chains in uncertain environments, Supply Chain Anal. 10 (2025) 100119, http://dx.doi.org/10.1016/j.sca.2025.100119.
[27] R. Genuer, J.-M. Poggi, C. Tuleau-Malot, N. Villa-Vialaneix, Random forests for big data, Big Data Res. 9 (2017) 28–46.

[28] P. Zhang, Y. Jia, Y. Shang, Research and application of xgboost in imbalanced data, Int. J. Distrib. Sens. Netw. 18 (6) (2022) 15501329221106935.
[29] Y. Yusoff, M.S. Ngadiman, A.M. Zain, Overview of NSGA-II for optimizing machining process parameters, Procedia Eng. 15 (2011) 3978–3983.
[30] A. Bazyar, M. Abbasi, Multi-objective planning for a multi-echelon supply chain using parameter-tuned meta-heuristics, Adv. Manag. Oper. Res. 7 (1) (2025) 45–65, http://dx.doi.org/10.35912/amor.v7i1.2542.
[31] Y. Li, An enhanced genetic-based multi-objective mathematical model for industrial supply chain network, PLoS One 20 (3) (2025) e0311887, http://dx.doi.org/10.1371/journal.pone.0311887, PMC11878948.
[32] C. Little, S. Choudhury, T. Hu, K. Salomaa, Comparison of genetic operators for the multiobjective pickup and delivery problem, Mathematics 10 (22) (2022) 4308, http://dx.doi.org/10.3390/math10224308.
[33] L. Zhao, J. Xie, Optimization of the supply chain network planning problem using an improved genetic algorithm, Int. J. Simul. Multidiscip. Des. Optim. 14 (2023) 9, http://dx.doi.org/10.1051/smdo/2023009.
[34] W. Li, T. Friedrich, T. Kötzing, F. Neumann, A crowding distance that provably solves the difficulties of NSGA-II in many-objective optimization, 2024, http://dx.doi.org/10.48550/arXiv.2407.17687, arXiv preprint arXiv:2407.17687.
[35] US Environmental Protection Agency, Emission factors for greenhouse gas inventories, 2022, URL https://www.epa.gov/climateleadership/center-corporate-climate-leadership-ghg-emission-factors-hub. (Accessed August 2025).
[36] OpenStreetMap Contributors, OpenStreetMap, 2024, URL https://www.openstreetmap.org. Open-source collaborative mapping project. (Accessed November 2025).
[37] Project OSRM, Open source routing machine (OSRM), 2024, URL http://project-osrm.org. High-performance routing engine for OpenStreetMap data. (Accessed November 2025).
[38] Administrador de Infraestructuras Ferroviarias, ADIF: Spanish railway infrastructure administrator network data, 2024, URL https://www.adif.es. Official Spanish railway infrastructure data. (Accessed November 2025).
[39] European Environment Agency, Rail and waterborne freight transport: Decarbonisation scenarios, 2021, URL https://www.eea.europa.eu/publications/rail-and-waterborne-freight. (Accessed November 2025).
[40] UK Department for Transport, Carbon savings from modal shift in freight transport, 2022, URL https://www.gov.uk/government/publications/freight-carbon-review. (Accessed November 2025).
[41] L.O.B. da Silva Santos, K. Burger, R. Kaliyaperumal, M.D. Wilkinson, FAIR data point: a FAIR-oriented approach for metadata publication, Data Intell. 5 (1) (2023) 163–183.
[42] S.K. Patel, D. Panday, Optimizing credit card fraud detection: A genetic algorithm approach with multiple feature selection methods, ADCAIJ: Adv. Distrib. Comput. Artif. Intell. J. 13 (2024) e31533–e31533.
[43] W. Chen, L. Zhang, Q. Liu, An 'internet of things' enabled dynamic optimization method for smart vehicles and logistics tasks towards green logistics, Transp. Res. E 147 (2024) 102234, http://dx.doi.org/10.1016/j.tre.2024.102234.
[44] J. Fichtinger, S. Ries, E.H. Grosse, C.H. Glock, Environmental impact of warehousing: A scenario analysis for the United States, Int. J. Prod. Econ. 267 (2024) 109089, http://dx.doi.org/10.1016/j.ijpe.2024.109089.
[45] E. Pekel, Deep learning approach to technician routing and scheduling problem, ADCAIJ: Adv. Distrib. Comput. Artif. Intell. J. 11 (2) (2022) 191.
[46] G.S. Balan, V.S. Kumar, S.A. Raj, Machine learning and artificial intelligence methods and applications for post-crisis supply chain resiliency and recovery, Supply Chain Anal. 10 (2025) 100137, http://dx.doi.org/10.1016/j.sca.2025.100137.